



Original Research

Co-transplantation of Schwann cells and neural stem cells in the laminin-chitosan-PLGA nerve conduit to repair the injured recurrent laryngeal nerve in SD rats

Yu Li ¹ · Yongzhi Men¹ · Baoxin Wang¹ · Xinwei Chen¹ · Ziwei Yu¹

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Abstract

The objective of this paper is to investigate the possibility and efficacy of recurrent laryngeal nerve repair by transplantation of co-cultured Schwann cells and neural stem cells (NSCs) in laminin-chitosan-poly-lactic-co-glycolic acid (laminin-chitosan-PLGA) nerve conduits in rats. A laminin-chitosan-PLGA conduit was used in a rat recurrent laryngeal nerve transection model. The rat recurrent laryngeal nerve was dissected to generate a 5 mm defect. Then, a laminin-chitosan-PLGA nerve conduit with or without Schwann cells and NSCs in the lumen was transplanted into the defect. A total of 96 female rats were randomised into six groups: co-culture of NSCs and Schwann cells in the nerve conduit group (CO), Schwann cells only in the nerve conduit group (SC), neural stem cells only in the nerve conduit group (NSC-only), nerve conduit group (null), autologous nerve graft group (autograft) and sham operation group (sham). Regenerated nerves were evaluated by histological and functional assessment at 8 and 12 weeks after surgery. The diameter and area of the regenerated myelin sheath, as well as the secretion of brain-derived neurotrophic factor and glial cell-derived neurotrophic factor in laryngeal muscle or regenerated nerve tissue in the CO group, were significantly better than they were in the SC, NSC-only and null groups (all P values < 0.05). Immunofluorescence showed that the CO group had significantly more neurofilament-200 immunoreactive and S-100 immunoreactive fibres than the SC, NSC-only and null groups (all P values < 0.05). The performance of the CO groups and autograft groups was found to be similar by laryngoscopy. Arytenoid cartilage motion recovery in these two groups was significantly better than it was in the other groups (all P values < 0.05). Our results indicated that co-culture of Schwann cells and NSCs in laminin-chitosan-PLGA conduits might promote injured nerve regeneration. This method might be a promising alternative for defective nerve repair.

1 Introduction

Peripheral nerve injuries are global clinical problem that are quite different from other tissue damage [1]. These injuries greatly affect a patient's quality of life and are associated with a substantial social and economic burden. The peripheral nerve has strong inbuilt repair mechanisms. For example, when the two ends of the damaged nerve are

re-connected, the nerve is likely to recover most of its functions. In the field of head and neck surgery, damage to the recurrent laryngeal nerve is one of the most common surgical complications, with an incidence of 0.3–3%, even in the most experienced tertiary hospital [2, 3]. Severe subsequent outcomes may occur after the damage, including irreversible hoarseness and lethal dyspnea [4]. Therefore, a critical topic in neural regeneration is the recurrent laryngeal nerve after its damage.

Currently, autologous nerve transplantation has been widely performed in the clinic [5]. However, such a procedure may cause subsequent complications of sensory and functional disturbances at the donor site, such as traumatic neuroma [6–8]. To address these problems, tissue engineering technology has been widely applied with the combined use of seed cells, nerve conduits and growth factors in the repair of peripheral nerve injury.

Preliminary data from our previous study showed that in comparison to traditional methods, there were better

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✉ Ziwei Yu
david545yu@163.com

¹ Department of Otolaryngology Head and Neck Surgery, Shanghai General Hospital, Shanghai Jiao Tong University School of Medicine, Shanghai 200080, PR China

outcomes of neural regeneration using a novel technique of co-culturing neural stem cells (NSCs) and Schwann cells (SCs) in a laminin-chitosan-poly-lactic-co-glycolic acid (laminin-chitosan-PLGA) nerve conduit [9].

The results indicated that co-cultured SCs and NSCs might promote the differentiation of NSCs into neurons [10]. In addition, compared to other methods the use of laminin and chitosan in the conduits might increase the conglutination of the cells, promote cell growth, provide better biocompatibility and improve the microenvironment for regeneration [11, 12].

In the present study, our objective is to provide additional evidence of utilising the technique to repair laryngeal recurrent nerves for clinical translational prospects.

2 Materials and methods

2.1 Experimental animals

Female Sprague Dawley rats (180) weighing ~150 g were randomly divided into six groups (Table 1): co-culture group (CO group, laminin-chitosan-PLGA nerve conduit + SC + NSC, $n = 30$), Schwann cell group (SC group, laminin-chitosan-PLGA nerve conduit + SC, $n = 30$), neural stem cell only group (NSC-only group, laminin-chitosan-PLGA nerve conduit + NSC, $n = 30$), null group (laminin-chitosan-PLGA nerve conduit, $n = 30$), autograft group ($n = 30$) and sham group ($n = 30$). All rats were provided standard mouse chow and water ad libitum, and they were housed under a 12-h light/dark cycle.

The study was approved by the Institutional Review Board of Shanghai General People's Hospital, Shanghai, PR China.

2.2 Cell culture

2.2.1 Culture of Schwann cells

SCs were obtained from the sciatic nerves of experimental rats. The culture of SCs in this study was performed based

on traditional methods with improvements, which included the following key aspects: (1) experimental animals were selected from 3 to 5-day newborn rats; (2) as much of the outer membrane of the nerve and connective tissue was stripped off as possible; (3) the time of enzyme digestion was prolonged; and (4) forskolin and bFGF were added to the SC culture medium.

2.2.2 Culture of neural stem cells

The uteri of pregnant SPF 14–16-day-old rats were exposed after they were humanely killed. The cerebral hemispheres were split and stripped of obvious vasculature, and the hippocampal tissue was isolated. Hippocampal tissue was transferred to 3.5 mm culture dishes that contained DMEM/F12. The cell suspension was then centrifuged. Then, the resulting supernatant was discarded. The precipitate was resuspended in NSCM, and the number of cells was counted. Subsequently, the cells were plated in a T25 flask at a density of $1 \times 10^6/\text{mL}$, and then they were cultured in an incubator at 37 °C (5% CO₂). The cells were passaged every 7 days. The NSCs were identified by nestin staining thereafter (BD Pharmingen; BD Biosciences, Franklin Lakes, NJ, USA).

2.3 Nerve conduit

New nerve conduits were supplied by the Department of Materials, Donghua University, Shanghai, PR China. They were fabricated using braided PLGA material that was coated with laminin and chitosan. Compared to uncoated conduits, laminin and chitosan in the conduits increase the conglutination of the cells, promote cell growth, provide better biocompatibility and improve the microenvironment for regeneration [13–16]. The parameters of the nerve conduit were as follows: inner diameter, 0.6 mm; thickness of tube wall, 0.2 mm; and length, 2.1 cm. Built into the conduit were ten nanofibre filaments. Detailed information on the development of the nerve conduit has been described elsewhere [9].

2.4 Surgical procedures

Surgery was performed under deep anaesthesia, which was induced by an injection of 5% chloral hydrate (0.7 mL/100 g bodyweight). Experimental animals were placed on the operating table, and an incision was made in the skin and fascia. The muscles were separated, and the larynx and trachea were exposed. Subsequently, a 7 mm recurrent laryngeal nerve segment, located in the tracheoesophageal groove, was excised.

The treatment of each group was as follows:

Table 1 Different experimental groups in the present study

Groups	No. of rats	Tissues		
		Laminin-chitosan-PLGA nerve conduit	NSC	SC
CO group	30	Yes	Yes	Yes
SC group	30	Yes	No	Yes
NSC-only group	30	Yes	Yes	No
Null group	30	Yes	No	No
Autograft group	30	n/a		
Sham group	30	n/a		

CO group: 1 mm of the proximal stump was sutured into a 5 mm long laminin-chitosan-PLGA nerve conduit, and then 22.5 μL Matrigel, 3.75 μL SC (at least 0.083×10^6 cells) and 3.75 μL NSC ($\sim 0.083 \times 10^6$ cells) were injected into the conduit. The distal stump was sutured to the other end of the conduit, leaving a 5 mm gap between the stumps.

SC group: the approach was similar to that of the CO group, but the injection mixture was 22.5 μL Matrigel and 7.5 μL SC ($\sim 0.167 \times 10^6$ cells).

NSC-only group: the approach was similar to that of the CO group, but the injection mixture was 22.5 μL Matrigel and 7.5 μL NSC ($\sim 0.167 \times 10^6$ cells).

Null group: the approach was similar to that of the CO group, but the injection contained only 30 μL Matrigel.

Autograft group: the recurrent laryngeal nerve (RLN) was separated from tracheoesophageal groove, the proximal and distal ends of the 5-mm nerve segment were swapped, and their corresponding nerve adventitia sutured.

Sham group: the RLN was separated from the tracheoesophageal groove, but no intervention was necessary.

An electrolaryngoscope examination was performed after surgery, and the fixation of the bilateral arytenoid cartilage indicated the success of modelling.

2.5 Detection methods

2.5.1 Electron microscopy

Eight and 12 weeks after surgery, five rats were randomly selected from each group. Specimens were assessed by toluidine blue staining and transmission electron microscopy. Following administration of anaesthesia, the intermediate segment of nerve regeneration was excised and fixed in 2.5% glutaraldehyde and 1% osmium tetroxide. Subsequently, the tissue was dehydrated in graded ethanol and embedded in epoxy resin. The treated specimen was then cut into 1 mm thick sections. The epoxy-embedded tissue was cut into ultrathin sections and was followed by citrate staining (H7650, Hitachi, Tokyo, Japan), permitting the measurement of the diameter and area of the myelin sheath under an electron microscope. All images were processed by IPP Version 6.0 software.

2.5.2 Toluidine blue staining

The processing of the specimen for toluidine blue staining was the same as it was for electron microscopy. The epoxy-embedded tissue was cut into ultrathin sections (50 nm) and then observed under a microscope after 1% toluidine blue staining; specimens were photographed using an AXI-OCAM digital camera (Zeiss, Germany). The number of regenerated myelinated nerve fibres in each image was counted (Supplementary Figure). As a gold standard of

peripheral nerve staining, myelin sheaths could be clearly identified by sharp delimitation [17].

2.5.3 Real-time qPCR

Eight and 12 weeks after surgery, five rats in each group were randomly selected for analysis by real-time qPCR detection. RNA was isolated from the regenerated nerves. The concentration of RNA was quantitated by OD values. The prepared RT reaction solution was mixed gently, and then reverse transcription was performed. Finally, the configured PCR solution was maintained on ice until it was used real-time PCR experiments. The primer sequences and other PCR information are presented in Supplementary Table.

The average reference and target gene CT values were collected. The ΔCt value can be calculated from the target gene CT minus the reference gene CT; then the $2^{-\Delta\text{Ct}}$ can be calculated, i.e., ΔCt . This provides the relative expression of each sample gene.

2.5.4 Electron laryngoscope

Eight and 12 weeks after surgery, all rats from each group were examined by electron laryngoscopy before other examinations were performed. An electroscope measurement tool was used to measure the diameter, thickness and area of myelin from five randomly chosen sites. The average of five measurements is presented as the result.

2.6 Statistical analysis

All data are expressed as the mean $\pm$ standard deviation (SD), and all statistical analyses were performed using the software Prism GraphPad version 6.0 (GraphPad Software, Inc., USA). An unpaired *t*-test was used for normally distributed data with an equal SD to test for differences between two samples in a group at various time points. A *P* value < 0.05 was considered to be statistically significant.

3 Results

All experimental animals survived without any complications, including infections. Surgical wounds healed well.

Figure 1 shows the semi-thin cross-sections of regenerating laryngeal nerves under electron microscopy in different groups 8 and 12 weeks after surgery. The CO group had a larger number of regenerated nerve fibres than the other groups. Thicker substantial myelin with less connective tissue was also observed in the CO group. Such regenerated myelin had matured well and exhibited a consistent thickness. The thickness of myelin sheaths was

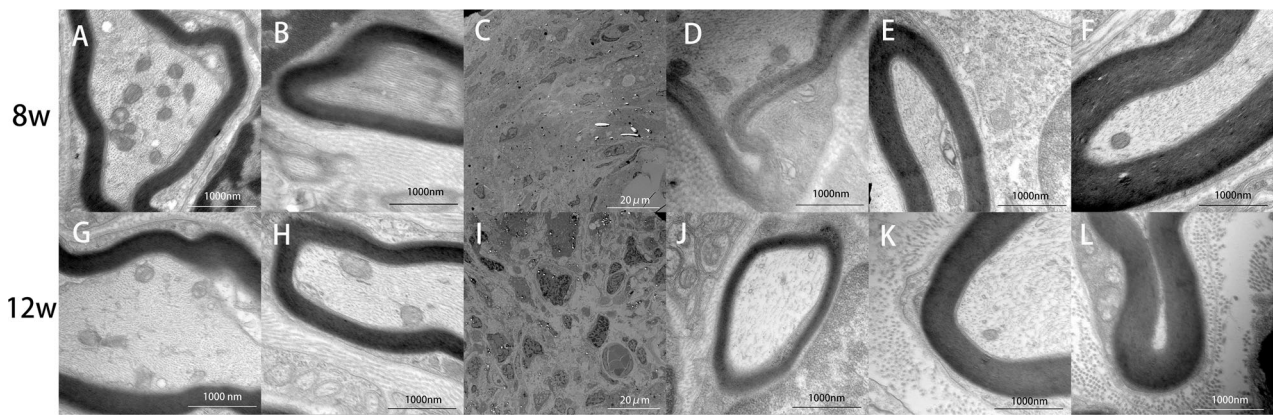


Fig. 1 Semi-thin cross-sections of regenerating laryngeal nerves from the CO, SC, NSC-only, null, autograft and sham groups, as assessed under electron microscopy at 8 (a–f) and 12 weeks (g–l) after surgery

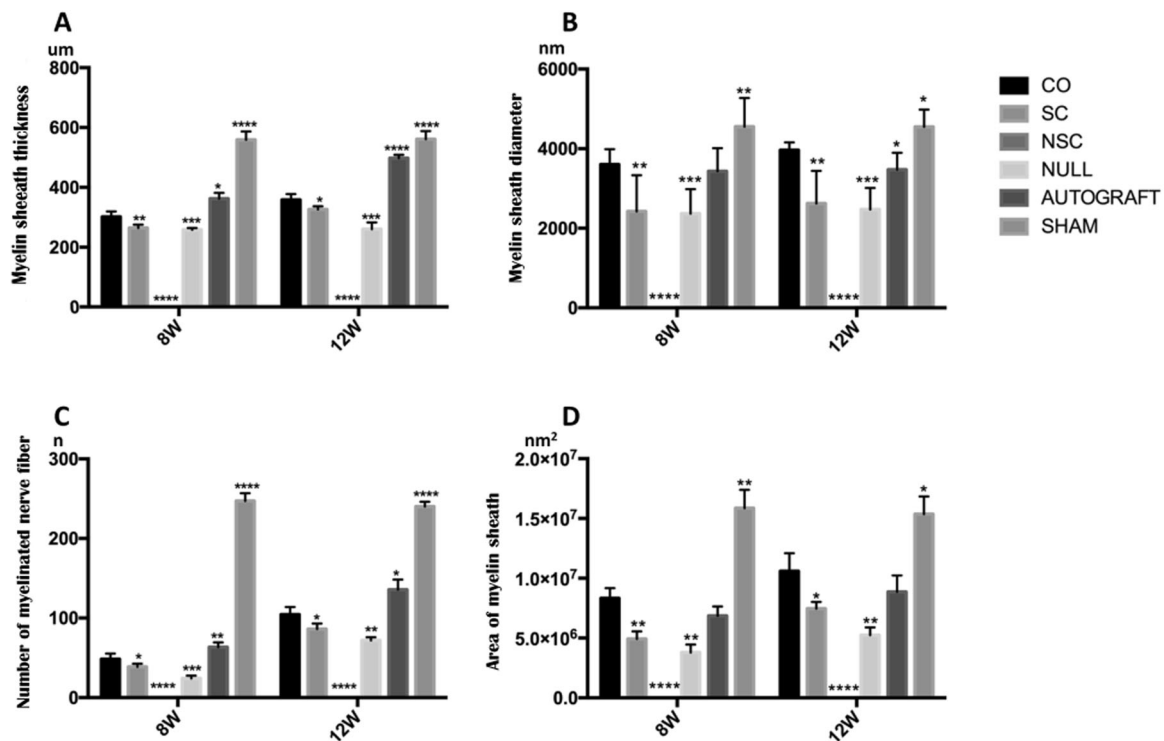


Fig. 2 The myelin sheath thickness (a), diameter (b), myelinated nerve fibre (c) and area of myelin sheaths (d) in the CO, SC, NSC-only, null, autograft and sham groups at 8 and 12 weeks after surgery. *P* value summary: **P* < 0.05, ***P* < 0.01, ****P* < 0.001 and *****P* < 0.0001

significantly larger in the CO group than in the SC and NSC-only groups (all *P* < 0.05, Fig. 2a). However, it was not as thick as then sheaths in the autograft and sham groups. Regenerated axons also developed well and were arranged in order in the CO group. A large quantity of connective tissue, inflammatory cells, fragmentation of nuclei and unabsorbed Matrigel inside the NSC-only group's conduit were observed. However, no significant amount of newly generated myelin was detected.

The diameter of myelin sheaths (Fig. 2b), as well as the number of myelinated nerve fibres (Fig. 2c), was significantly larger in the CO group than it was in the other groups at week 12 after surgery (all *P* < 0.05). No significant difference was observed in terms of the area of myelin sheaths 8 or 12 weeks after surgery when comparing the CO group and the autograft group (Fig. 2d).

Reverse-transcription PCR analysis of mRNA levels was then used to evaluate the difference in brain-derived

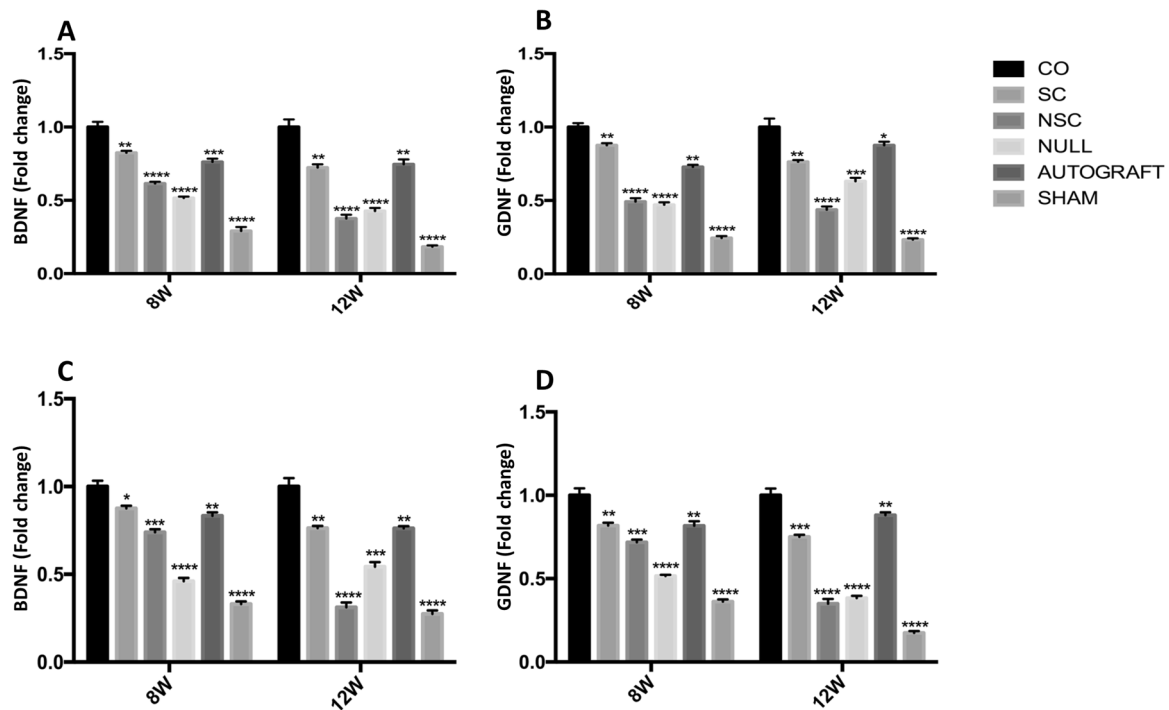


Fig. 3 The fold change in BDNF expression (a) and GDNF expression (b) in laryngeal muscles and the fold change in BDNF expression (c) and GDNF expression (d) in regenerated recurrent laryngeal nerves in

the CO, SC, NSC-only, null, autograft and sham groups at 8 and 12 weeks after surgery. *P* value summary: **P* < 0.05, ***P* < 0.01, ****P* < 0.001 and *****P* < 0.0001

neurotrophic factor (BDNF) and glial cell-derived neurotrophic factor (GDNF) expression in laryngeal muscles and nerves (Fig. 3a–d). Eight and 12 weeks after surgery, the expression of BDNF and GDNF in both laryngeal muscles and nerves was significantly higher in the CO group than it was in the other groups (all *P* < 0.05).

Figure 4 shows the open and closed state of arytenoid cartilage. Eight weeks after surgery, abduction on the surgical side of the arytenoid cartilage was weaker and slightly delayed compared to the contralateral side in the CO group (Fig. 4a: a, g). The glottis could not be completely closed. After 12 weeks, the abduction ability on the surgical side of the arytenoid cartilage had recovered, there was no contradictory movement, and the glottis was closed (Fig. 4b: a, g). The performance of the autograft (Fig. 4a: e, k, b: e, k) and CO groups was similar. These results indicated that the function of the regenerated nerves in the CO group was similar to that in the autograft group. Comparing the difference in changes between the maximum and the minimum of the movement of vocal cords among groups, the average change was significantly larger in the CO group than in the SC, NSC-only groups and null group (all *P* < 0.05, Fig. 5). However, the change in movement was smaller than that what was observed in the sham group. These results indicated that there was better functional recovery in the CO group than there was in the other groups.

It was observed in the SC group at week 8 that the surgical side of the arytenoid cartilage was fixed (Fig. 4a: b, h). Twelve weeks after surgery, the abduction of the right arytenoid cartilage was worse (Fig. 4b: b, h). Compared with the contralateral side, the abduction was delayed, and the glottis could close. The right side of the arytenoid cartilage was fixed next to the middle position in the NSC-only group at both weeks 8 and 12 (Fig. 4a: c, i, b: c, i), but the left side could compensate. However, the glottis could not be completely closed. The arytenoid cartilage on the right side in the null group 8 weeks after surgery was fixed to the paramedian position; the left side could compensate, but the glottis could not be completely closed. After 12 weeks, the abduction on the right arytenoid cartilage was slightly weak, and, again, the glottis could not be completely closed (Fig. 4a: d, j, b: d, j). These results suggested that recovery of neural function was better in the CO group than in the SC or NSC-only groups (Fig. 5).

4 Discussion

In the present study, we were able to obtain additional evidence to confirm that co-cultured SCs and NSCs could promote the differentiation of NSCs into neurons and therefore improve the recovery of neural function after damage to the



Fig. 4 At 8 (a) and 12 weeks (b) after surgery, the largest open state of arytenoid cartilage (a–f) and the minimum closed state of arytenoid cartilage (g–l) were observed in the CO, SC, NSC-only, null, autograft and sham groups

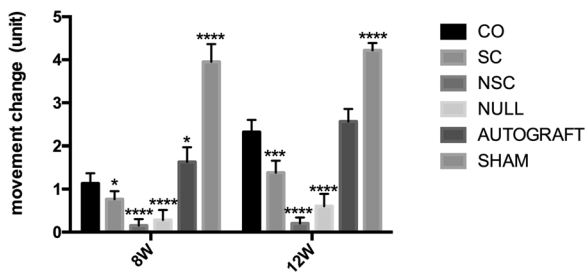


Fig. 5 The movement change of vocal cords (changes between the maximum and the minimum of the movement of vocal cords) in the CO, SC, NSC-only, null, autograft and sham groups at 8 and 12 weeks after surgery. *P* value summary: **P* < 0.05, ***P* < 0.01, ****P* < 0.001 and *****P* < 0.0001

recurrent laryngeal nerve. Such a technique may be utilised in clinical practice after comprehensive evaluation of safety and efficacy in human subjects in the future.

The chitosan-PLGA composite in the present study was prepared by the combination of PLGA material, chitosan and laminin, and the laminin-chitosan-PLGA composite conduit was optimised through coating modification. To the best of our knowledge, this technique was first reported

by our group. Such material was reported to have better biocompatibility than other materials and exhibited no cytotoxicity [13–16]. The vertical nanofibres in the conduit (that were alongside the axial direction of the conduit) can guide the direction of axon growth. When SCs were co-cultured with NSCs in vitro in the conduit, a better growth status and longer neurites resulted compared to what was observed with other treatments. Most NSCs differentiated into neurons in previous and present studies, neurotrophic factors (BDNF and GDNF) were secreted, and the cell survival rate was significantly higher than what was observed in cells cultured alone. NSC and SC co-culture may also improve the nerve regeneration micro-environment [18]. Therefore, we concluded that the laminin-chitosan-PLGA nerve conduit co-transplanted with SC and NSC has the ability to promote the repair of recurrent laryngeal nerve defects.

The ultrastructure of the regenerative nerve is an important indicator of the degree of nerve maturation, and the diameter of the regenerated nerve fibre and the thickness of the myelin sheath directly reflect the maturity of the axon and myelin sheath. In this study, we observed a large number of regenerated nerve fibres and a thicker myelin

sheath in the CO group than we did in other groups, indicating an improvement in maturity of neural regeneration.

A large number of fibroblasts, inflammatory cells and fragmented nuclei were observed in the NSC-only group; however, no neuron regeneration was observed. Similar results were reported by other groups. Some studies have suggested that the survival rate of NSC transplantation alone *in vivo* is low [19]. Yihua et al. indicated that factors secreted by SCs can substantially support the growth of NSCs; if NSCs lose the support of SCs, the NSCs will gradually die [20]. These results might suggest that the microenvironment could be one of the most critical factors for NSC survival and differentiation. Further investigation regarding the mechanism of the interaction between the microenvironment and neural regeneration should be conducted.

Although the current study achieved promising outcomes, several problems should be noted. First, the side effects of cell-based therapies have been widely discussed and reported [21–23], including immune rejection of allogeneic stem cells, inflammation, connective tissue scars, intestinal flora disturbance and even tumorigenesis. These problems have been found when using embryonic and adult stem cells to repair nerve injuries. In addition, NSCs should be used after predifferentiation to reduce the incidence of tumours. Immune rejection, tumorigenesis and other adverse events were not found in the present study; however, this might be due to the relatively short observation period. Such adverse events should be fully evaluated in future studies. Second, we did not perform a degradation study of the conduit or the mechanical strength evaluation of the conduit at this stage. The purpose of the present study was to provide evidence of neural regeneration using a system of SC and NSC co-culture in the conduit. Additional investigations regarding these topics should be conducted in the future before utilising such systems in clinical practice.

5 Conclusions

In conclusion, neural regeneration following SC and NSC co-culture in LC-PLGA conduits was significantly better than other methods, including traditional autologous nerve transplantation. This indicated that co-culture of Schwann cells and NSCs in laminin-chitosan-PLGA conduits might promote injured nerve regeneration, which might be a promising alternative method for defective nerve repair.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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